

A



DeepMutate-3D

Protein language model mutation scanning, painted onto the AlphaFold fold.

ESM-2 scores every possible point mutation as a log-likelihood ratio against the wildtype residue. Positions the model refuses to change are evolutionarily constrained, likely structural or functional cores, and glow **red**. Positions it happily swaps glow **blue**.

Example protein

Haemoglobin subunit alpha, P69905 (142 aa)

Load sequence

Protein sequence (FASTA or raw)

MVLSPADKTNVKAAGWKGAHAGEYGAEALERMF~~L~~SPPTTKTYFPHFDLSHGSAQVKGHGKKVADALTNAVAHVDDMPNALS~~DL~~HAHKLRVDPVNFKLLSHCLLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR

UniProt ID

P69905

Used for the AlphaFold structure, and to fetch the sequence.

Model

ESM-2 650M (most accurate)

Larger models score better but cost more compute.

Scoring mode

Masked marginals (accurate)

Wildtype marginals (fast)

Masked marginals runs one forward pass per residue.

Scan Mutations & Render 3D Fold

Sensitivity table (.csv)

Scored structure (.pdb)

Scan complete

Residues scored: 142 | Device: cuda | Mode: Masked marginals (accurate)

Model: facebook/esm2_t33_650M_UR50D | Inference time: 3.99 s (28.1 ms/residue)

Mean sensitivity: -7.703 | Most fragile: D95 (-12.90), F44 (-12.47), H88 (-12.29), G60 (-11.79), G26 (-11.74)

AlphaFold structure retrieved for P69905.

B

Residue Sensitivity Array

Pos	WT	Sensitivity	Verdict	Most damaging	Most tolerated
1	M	-6.283	Sensitive	M1W (-8.24)	M1A (-3.96)
2	V	-3.611	Tolerant	V2R (-7.66)	V2S (2.05)
3	L	-7.703	Sensitive	L3K (-9.83)	L3F (-4.58)
4	S	-7.637	Sensitive	S4W (-12.31)	S4T (-0.78)
5	P	-1.96	Tolerant	P5W (-7.88)	P5A (3.43)
6	A	-5.127	Sensitive	A6W (-9.96)	A6E (-0.59)
7	D	-10.465	Critical	D7W (-14.32)	D7E (-1.05)
8	K	-5.023	Sensitive	K8W (-9.15)	K8V (-1.21)
9	T	-3.982	Tolerant	T9W (-10.53)	T9A (2.13)
10	N	-1.165	Tolerant	N10W (-6.51)	N10A (2.64)
11	V	-8.676	Critical	V11H (-12.32)	V11I (1.13)
12	K	-4.754	Sensitive	K12W (-10.57)	K12Q (-0.94)
13	A	-5.882	Sensitive	A13W (-11.48)	A13S (-1.33)
14	A	-3.053	Tolerant	A14D (-7.08)	A14I (3.37)
15	W	-11.563	Critical	W15D (-15.44)	W15F (-4.46)

C

Structural Heatmap (AlphaFold DB)

H88

POSITION 88

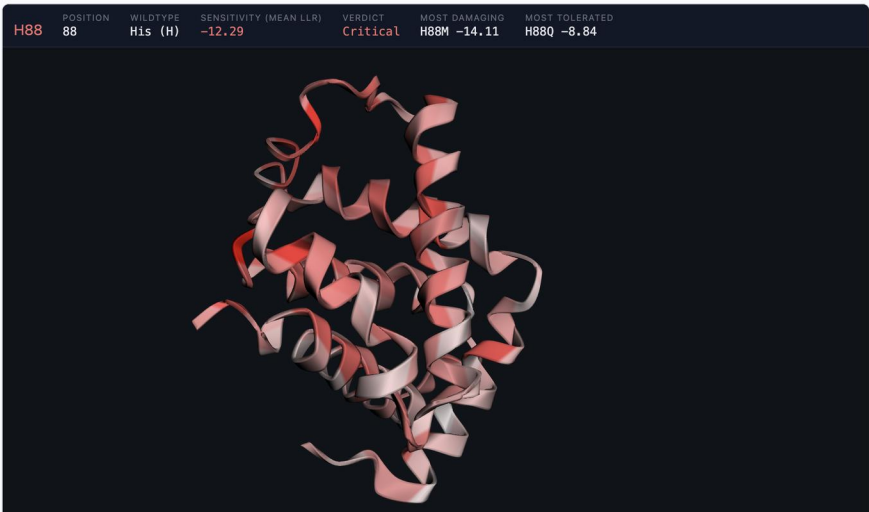
WILDTYPE His (H)

SENSITIVITY (MEAN LLR) -12.29

VERDICT Critical

MOST DAMAGING H88M -14.11

MOST TOLERATED H88Q -8.84



Sensitivity (mean LLR)

fragile -12.90

tolerant +12.90

- 142 residues painted